Roll No. | | | | | | | |

Name of the Course: M. Tech (**Geo-Informatics**) Semester: 3th

Name of the Paper: Application of Geo-Informatics for Natural resource management

Paper Code: MGI 313

Time: 1:30 Hours Maximum Marks: 50

Note:-

(i) All **the questions** are compulsory

(ii) Each question carries 10 marks

Q1	(10marks)	CO 1
(a)	Define land suitability analysis in the context of agriculture	
(b)	Discuss RS and GIS-based mapping of geomorphological characteristics of landforms.	
Q2	(10 marks)	CO 1
(a)	Explain how remote sensing supports watershed management.	
(b)	Illustrate how microwave RS is applied in estimating soil moisture for crop growth monitoring.	
Q3	(10 marks)	CO 2
(a)	Define soil moisture and its importance in agriculture	
(b)	Describe the role of GIS in integrating thematic layers for hazard zoning.	
Q4	(10 marks)	CO 2
(a)	Describe the role of satellite imagery in mapping forest cover change.	
(b)	Enlist the differences in wavelength coverage among panchromatic, multispectral, and hyperspectral sensors.	
Q5	(10 marks)	CO 2
(a)	Demonstrate the use of geospatial data for identifying biodiversity hotspots	
(b)	Explain how remote sensing supports crop health monitoring in precision agriculture	

25/9/25

III rd Sem

M. Tech Geo.

11 to 12:30